

Reviewer Pack — Tropical Hodge Norm Lower Bound for Resolution Proof Length

Entry c581f31d6b75 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Tropical Hodge Norm Lower Bound for Resolution Proof Length

Entry ID: c581f31d6b75 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 17:46:30 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry **Field B** (complexity object): Resolution Proof Complexity

Statement:

{‘text’: ‘For any polynomial-sized CNF formula F with n variables, the tropical Hodge norm of its associated tropical curve is lower-bounded by $\Theta(2^n / \log(n))$ in terms of resolution proof length. Equivalently: for a given number of steps $k \leq \log(n)$, there exists at least one CNF that requires at least $2^k / \log(n)$ steps to be resolved.’, ‘counterexample’: ‘Provide a CNF formula F with n variables requiring less than $2^n / \log(n)$ resolution steps, which would disprove the conjecture.’}

Rationale (proposer’s reasoning):

{‘text’: ‘Tropical geometry provides a bridge between algebraic geometry and computational complexity theory. The Hodge norm in tropical geometry is known to be related to the genus of curves, which can serve as a lower bound for computation costs in certain models. This conjecture suggests that such a relationship may also exist for resolution proofs, offering a potential new approach to circuit complexity.’}

Taxonomy category: Geometric Combinatorics (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 021bdcd7ed264116

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for a sample of random CNF formulas with n variables, at least 80% (support_fraction ≥ 0.8) have a resolution proof length ratio (tropical Hodge norm to $\log(n)$) that meets or exceeds $2^k / \log(n)$, where $k \leq \log(n)$. The conjecture is falsified if any seed produces a resolution proof length ratio less than $2^k / \log(n)$ for $k \leq \log(n)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.90	UNCERTAIN	HITS
KARP_LIPTON	SAFE	1.00	UNCERTAIN	SAFE

4. Novelty audit

Verdict: NOVEL against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - tropical geometry AND resolution proof complexity AND Hodge norm - resolution proof length lower bound CNF tropical geometry - CNF formula resolution steps lower bound tropical Hodge norm

Top relevant hits considered: - [<http://arxiv.org/abs/1406.3065v2>] Lower Bounds for Tropical Circuits and Dynamic Programs - [<http://arxiv.org/abs/1207.2443v2>] Tropical Teichmuller and Siegel spaces - [<http://arxiv.org/abs/1204.3875v2>] Tropicalizing vs Compactifying the Torelli morphism

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=102.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def gaussian_elimination(A):
        m, n = len(A), len(A[0])
        for i in range(m):
            max_row = i
            for j in range(i+1, m):
                if abs(A[j][i]) > abs(A[max_row][i]):
                    max_row = j
            A[i], A[max_row] = A[max_row], A[i]
            for j in range(i+1, m):
                factor = A[j][i] / A[i][i]
                for k in range(n):
                    A[j][k] -= factor * A[i][k]
        return A

    def matrix_multiply(A, B):
        m, n, p = len(A), len(B), len(B[0])
        C = [[0]*p for _ in range(m)]
        for i in range(m):
            for j in range(p):
                for k in range(n):
```

```

        C[i][j] += A[i][k] * B[k][j]
    return C

def determinant(A):
    m, n = len(A), len(A[0])
    if m != n:
        raise ValueError("Matrix must be square")
    if n == 1:
        return A[0][0]
    det = 0
    for j in range(n):
        submatrix = [row[:j] + row[j+1:] for row in A[1:]]
        det += (-1)**j * A[0][j] * determinant(submatrix)
    return det

def tropical_hodge_norm(A):
    m, n = len(A), len(A[0])
    max_row = [max(row) for row in A]
    min_col = [min(col) for col in zip(*A)]
    return sum(max_row[i] + min_col[i] for i in range(m))

def resolution_length(F):
    # Simplified DPLL solver to estimate resolution length
    stack = []
    literals = set()
    for clause in F:
        literals.update(clause)
    while literals:
        literal = random.choice(list(literals))
        literals.remove(literal)
        stack.append((literal, True))
        stack.append((literal, False))
    return len(stack)

n = 10
F = []
for _ in range(2**n):
    clause = [random.randint(-n, n) for _ in range(n)]
    if all(x != 0 for x in clause):
        F.append(clause)

H = [[random.randint(-n, n) for _ in range(n)] for _ in range(n)]
det_H = determinant(H)
norm_H = tropical_hodge_norm(H)

k = math.ceil(math.log2(n))
length = resolution_length(F)

metric_value = norm_H / math.log(n)
conjecture_holds = metric_value >= 2**k / math.log(n)
counterexample = "" if conjecture_holds else "mapping_undefined"

return {
    "metric_name": "Tropical Hodge Norm Ratio",
    "metric_value": metric_value,
    "instances_tested": len(F),

```

```

        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or [2**i + 1 for i in range(5, 8)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(i for i, r in enumerate(results) if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

Ratio', 'metric_value': -1.3028834457097553, 'instances_tested': 597, 'conjecture_holds': False, 'co
TRIAL: {'metric_name': 'Tropical Hodge Norm Ratio', 'metric_value': 4.3429448190325175, 'instances_t
TRIAL: {'metric_name': 'Tropical Hodge Norm Ratio', 'metric_value': -3.908650337129266, 'instances_t
TRIAL: {'metric_name': 'Tropical Hodge Norm Ratio', 'metric_value': -0.43429448190325176, 'instances
TRIAL: {'metric_name': 'Tropical Hodge Norm Ratio', 'metric_value': 3.0400613733227626, 'instances_t
TRIAL: {'metric_name': 'Tropical Hodge Norm Ratio', 'metric_value': 4.777239300935769, 'instances_te
TRIAL: {'metric_name': 'Tropical Hodge Norm Ratio', 'metric_value': 2.6057668914195107, 'instances_t
TRIAL: {'metric_name': 'Tropical Hodge Norm Ratio', 'metric_value': 3.908650337129266, 'instances_te
TRIAL: {'metric_name': 'Tropical Hodge Norm Ratio', 'metric_value': -3.474355855226014, 'instances_t
TRIAL: {'metric_name': 'Tropical Hodge Norm Ratio', 'metric_value': 3.0400613733227626, 'instances_t
TRIAL: {'metric_name': 'Tropical Hodge Norm Ratio', 'metric_value': -1.737177927613007, 'instances_t
RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed=0

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test reports a counterexample with ‘mapping_undefined’, which suggests a potential bug in the metric definition or construction of the tropical Hodge norm, rather than an actual counterexample to the conjecture.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results indicate that for at least one seed, the resolution proof length ratio was less than $2^k / \log(n)$ for some $k \leq \log(n)$, which meets th | next: Investigate the ‘mapping_undefined’ error to determine if it is a genuine counterexample or an issue with the metric definition. If it is a genuine counterexample, further analysis is needed to understand the properties of the CNF formula that leads to this result.

11. Audit log (LLM calls)

Total LLM calls: 12

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12070
2	propose	ollama_remote	glm4:latest	0	0	11323
3	propose	ollama_remote	glm4:latest	0	0	12815
4	preregistration	ollama_remote	glm4:latest	0	0	6455
5	novelty	ollama_remote	glm4:latest	0	0	4599
6	novelty	ollama_remote	glm4:latest	0	0	5499
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	40063
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8771
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15230
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18064
11	critic	ollama_remote	glm4:latest	0	0	42077
12	judge	ollama_remote	glm4:latest	0	0	6215

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 183180 ms total latency. Provider mix: {'ollama_remote': 12}

(full prompt+response transcripts available in `research/audit/c581f31d6b75.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/c581f31d6b75.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/c581f31d6b75.tar.gz` (if generated)